

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Hour QUESTIONS: 5

Time Duration: 1

Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I CHAKRADHAR PEELA (192412400)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Chakradhar Peela

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is

proved. **Question 2 – Student Assignment**

Submission Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is

proved. **Question 3 – Library**

Membership Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into

CNF. Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement

Eligibility Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm. **Tasks**

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.

2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

QUESTION 1 – RAIN AND WET GROUND

Knowledge Base

The given knowledge base contains two statements:

1. If it rains, then the ground becomes wet.
2. It is raining.

The goal is to prove that **the ground is wet**.

Step 1: Define Propositional Symbols

Let:

- **R** = It is raining.
- **W** = The ground is wet.

The first statement says:

If it is raining, then the ground is wet.

Therefore:

$$\mathbf{R \rightarrow W}$$

The second statement says:

It is raining.

Therefore:

$$\mathbf{R}$$

The goal is:

$$\mathbf{W}$$

Step 2: Convert the Knowledge Base into CNF

For resolution, implications must be converted into clauses.

We have:

$$\mathbf{R \rightarrow W}$$

Using the implication rule:

$$\mathbf{P \rightarrow Q \equiv \neg P \vee Q}$$

Therefore:

$$\mathbf{R \rightarrow W \equiv \neg R \vee W}$$

So the first CNF clause is:

$$\mathbf{C1 = \neg R \vee W}$$

The second statement is already a literal:

$$\mathbf{C2 = R}$$

Thus, the complete knowledge base in CNF is:

$$\mathbf{C1: \neg R \vee W}$$

$$\mathbf{C2: R}$$

Step 3: Negate the Goal

The goal is:

$$\mathbf{W}$$

For resolution by contradiction, we assume that the goal is false.

Therefore:

$$\mathbf{\neg W}$$

Add this as another clause:

$$\mathbf{C3 = \neg W}$$

Now our clauses are:

$$\mathbf{C1: \neg R \vee W}$$

$$\mathbf{C2: R}$$

$$\mathbf{C3: \neg W}$$

Step 4: Apply Resolution

Resolution 1

Take:

$$\mathbf{C1: \neg R \vee W}$$

$$\mathbf{C2: R}$$

The complementary literals are **$\neg R$** and **R** .

Remove them and combine the remaining literals:

$$\mathbf{W}$$

Therefore:

$$\mathbf{C4 = W}$$

Resolution 2

Now take:

C4: W

C3: $\neg W$

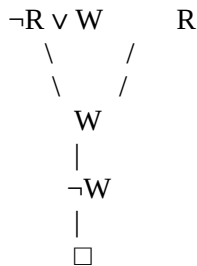
The complementary literals are **W** and **$\neg W$** .

Resolving them produces:

□

The symbol □ represents the **empty clause**.

Resolution Tree



Explanation

The original knowledge base tells us that whenever it rains, the ground becomes wet. We are also given that it is currently raining. Therefore, logically, the ground must be wet.

To prove this using resolution, we temporarily assume that the ground is **not wet**. The knowledge base derives **W** , while the negated goal gives **$\neg W$** . These two clauses contradict each other and produce the empty clause.

Final Conclusion

Since the resolution process produces the **empty clause** □, the negation of the goal leads to a contradiction.

Therefore, the goal is logically proved:

Hence, the ground is wet.

QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

Knowledge Base

The given knowledge base contains:

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

The goal is to prove that **Rahul receives internal marks**

Step 1: Define Propositional Symbols

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

The first statement becomes:

$S \rightarrow M$

The second statement becomes:

S

The goal is:

M

Step 2: Convert into CNF

The implication is:

$S \rightarrow M$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

we get:

$\neg S \vee M$

Therefore:

C1: $\neg S \vee M$

C2: S

These are the CNF clauses.

Step 3: Negate the Goal

The goal is:

M

Negation of the goal:

$\neg M$

Therefore:

C3: $\neg M$

Our complete set of clauses is:

C1: $\neg S \vee M$

C2: S

C3: $\neg M$

Step 4: Resolution

Resolution 1

Resolve:

C1: $\neg S \vee M$

C2: S

The complementary pair is:

$\neg S$ and S

After removing the complementary literals, we obtain:

M

Therefore:

C4 = M

Resolution 2

Now resolve:

C4: M

C3: $\neg M$

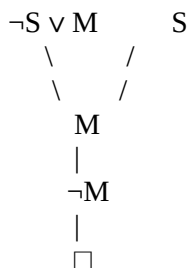
The complementary literals are:

M and $\neg M$

Therefore:

C5 = $\square$

Resolution Tree



Explanation

The first rule establishes a relationship between assignment submission and internal marks. Rahul has submitted the assignment. Therefore, the rule allows us to infer that Rahul receives internal marks.

For resolution, we negate the goal **M** and assume **$\neg M$** . However, the knowledge base produces **M** . Therefore, both **M** and **$\neg M$** exist, creating a contradiction.

The contradiction is represented by the empty clause $\square$.

Final Conclusion

Since the empty clause is obtained:

Rahul receives internal marks.

Therefore, the goal is successfully proved using the **Resolution Algorithm**.

QUESTION 3 – LIBRARY MEMBERSHIP

Knowledge Base

The knowledge base states:

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

The goal is to prove that **Priya can borrow books**.

Step 1: Propositional Logic Representation

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Statement 1:

$L \rightarrow B$

Statement 2:

L

Goal:

B

Step 2: CNF Conversion

We know:

L $\rightarrow$ **B**

Using:

P $\rightarrow$ **Q** $\equiv$ $\neg$ **P** $\vee$ **Q**

we get:

$\neg$ L $\vee$ **B**

Therefore:

C1: **$\neg$ L** $\vee$ **B**

C2: **L**

Step 3: Negate the Goal

The goal is:

B

Negating it:

$\neg$ B

Therefore:

C3: **$\neg$ B**

The complete clause set is:

C1: **$\neg$ L** $\vee$ **B**

C2: **L**

C3: **$\neg$ B**

Step 4: Resolution Process

First Resolution

Resolve:

C1: **$\neg$ L** $\vee$ **B**

C2: **L**

The complementary literals are **$\neg$ L** and **L**.

Therefore:

C4: **B**

Second Resolution

Now resolve:

C4: **B**

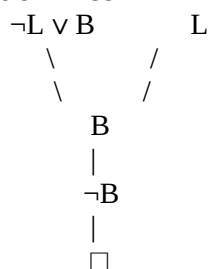
C3: **$\neg$ B**

The complementary literals are **B** and **$\neg$ B**.

Therefore:

C5: $\square$

Resolution Tree



Explanation

The knowledge base establishes that library membership is sufficient for borrowing books. Priya is explicitly stated to be a library member. Therefore, the rule can be applied to Priya.

In the resolution approach, we assume the opposite of the goal, which is **$\neg$ B**. The knowledge base derives **B**, while the negated goal provides **$\neg$ B**. The contradiction between these two clauses results in the empty clause.

Final Conclusion

Since:

B and **$\neg$ B** $\rightarrow$ $\square$

the goal is proved.

Therefore, Priya can borrow books.

QUESTION 4 – PLACEMENT ELIGIBILITY

Knowledge Base

The knowledge base contains:

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

The goal is to prove that **Arun is eligible for placement.**

Step 1: Propositional Representation

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

Statement 1:

A $\rightarrow$ **E**

Statement 2:

A

Goal:

E

Step 2: CNF Conversion

Convert:

A $\rightarrow$ **E**

Using:

P $\rightarrow$ **Q** $\equiv$ $\neg$ **P** $\vee$ **Q**

we obtain:

$\neg$ **A** $\vee$ **E**

Therefore:

C1: $\neg$ **A** $\vee$ **E**

C2: **A**

Step 3: Negate the Goal

The goal is:

E

Negated goal:

$\neg$ **E**

Therefore:

C3: $\neg$ **E**

Complete clause set:

C1: $\neg$ **A** $\vee$ **E**

C2: **A**

C3: $\neg$ **E**

Step 4: First Resolution

Resolve C1 and C2:

C1: $\neg$ **A** $\vee$ **E**

C2: **A**

C4: **E**

Here $\neg$ **A** and **A** are complementary literals.

Therefore, we derive:

E

Step 5: Second Resolution

Now resolve:

C4: **E**

C3: $\neg$ **E**

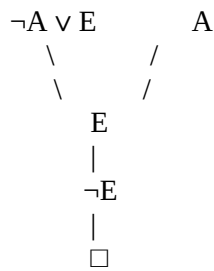
The complementary literals are:

E and $\neg$ **E**

Therefore:

C5: $\square$

Resolution Tree



Detailed Explanation

The first rule says that clearing the aptitude test is a condition for placement eligibility. Arun is explicitly given as having cleared the aptitude test.

Therefore, from:

$A \rightarrow E$

and:

A

we can derive:

E

For resolution, the goal E is negated to obtain $\neg E$. When the derived clause E is resolved with $\neg E$, a contradiction occurs.

The contradiction results in the empty clause $\square$.

Final Conclusion

The empty clause proves that the negation of the goal is inconsistent with the knowledge base.

Therefore, Arun is eligible for placement.

QUESTION 5 – ACCESS CONTROL SYSTEM

This is the **longest question** because it requires two implications and therefore requires multiple resolution steps.

The assessment specifically asks to convert all implications into CNF, negate the goal, derive the empty clause, and explain each resolution step.

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Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that:

The user is granted access.

Step 1: Define Propositional Symbols

Let:

- P = User entered the correct password.
- A = User is authenticated.
- G = User is granted access.

The three statements become:

Statement 1

If the user enters the correct password, then the user is authenticated.

$P \rightarrow A$

Statement 2

If the user is authenticated, then the user is granted access.

$A \rightarrow G$

Statement 3

The user entered the correct password.

P

Goal

G

Step 2: Convert All Statements into CNF

First implication

We have:

$P \rightarrow A$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

we get:

$$\neg P \vee A$$

Therefore:

$$C1 = \neg P \vee A$$

Second implication

We have:

$$A \rightarrow G$$

Using the same rule:

$$A \rightarrow G \equiv \neg A \vee G$$

Therefore:

$$C2 = \neg A \vee G$$

Third statement

The user entered the correct password:

$$P$$

Therefore:

$$C3 = P$$

So the CNF clauses are:

$$C1: \neg P \vee A$$

$$C2: \neg A \vee G$$

$$C3: P$$

Step 3: Negate the Goal

The goal is:

$$G$$

For resolution by contradiction, negate the goal:

$$\neg G$$

Therefore:

$$C4 = \neg G$$

The complete clause set becomes:

$$C1: \neg P \vee A$$

$$C2: \neg A \vee G$$

$$C3: P$$

$$C4: \neg G$$

Step 4: Apply Resolution

Resolution Step 1

Take:

$$C1: \neg P \vee A$$

$$C3: P$$

The complementary literals are:

$$\neg P \text{ and } P$$

Remove them.

The remaining literal is:

$$A$$

Therefore:

$$C5: A$$

This means that the user is authenticated.

Resolution Step 2

Now take:

$$C2: \neg A \vee G$$

$$C5: A$$

The complementary literals are:

$$\neg A \text{ and } A$$

After resolution, the remaining literal is:

$$G$$

Therefore:

$$C6: G$$

This means that the user is granted access.

Resolution Step 3

Now take:

C6: G

C4: $\neg G$

The complementary literals are:

G and **$\neg G$** .

Therefore:

C7: $\square$

The empty clause has been obtained.

Complete Resolution Derivation

C1: $\neg P \vee A$

C3: P

C5: A

Then:

C2: $\neg A \vee G$

C5: A

C6: G

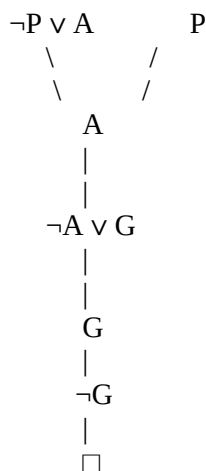
Finally:

C6: G

C4: $\neg G$

C7: $\square$

Resolution Tree



Detailed Explanation

The first rule establishes that entering the correct password results in authentication.

The third fact tells us that the user has entered the correct password.

Therefore, using resolution:

$P + (\neg P \vee A) \rightarrow A$

So, the user is authenticated.

The second rule states that an authenticated user is granted access:

$A \rightarrow G$

After CNF conversion:

$\neg A \vee G$

Since we have already derived **A**, we can resolve:

$A + (\neg A \vee G) \rightarrow G$

Therefore, the user is granted access.

However, to formally prove the goal using resolution, we initially added the negated goal:

$\neg G$

Now we have both:

G

and:

¬G

Resolving these complementary literals produces:

□

The empty clause indicates a contradiction.

Therefore, the assumption that the user is **not granted access** is false.

Final Conclusion

Since the resolution process successfully derives the **empty clause** □, the goal is logically entailed by the knowledge base.

Therefore, the user is granted access.